



CONTACT

Nikolas Pattakos

Software engineer

France

+33 6 61 60 97 53

+30 69 45 15 17 20

pattakosn@fastmail.com

SKILLS

C / C++ 15+ years

Linux & kernel 10+ years

HPC / Parallel (OpenMP, CUDA, MPI) 10+ years

Qt / GUI 7 years

Embedded / Yocto 4+ years

Agile / SCRUM / SAFe 7+ years

Languages: C, C++, Python, Bash, Fortran

Frameworks: Qt 5/6, OpenMP, CUDA, MPI

Tools: CMake, Git, GitLab, Jenkins, gdb, valgrind, perf, gprof

Libraries: BLAS / LAPACK, Eigen, Intel MKL

Domains: Numerical simulation, 3D / CAD / CAE, Linux/embedded, GPGPU

LANGUAGES

Greek: Native

English: Fluent

French: DELF B2

PROFILE

Senior C++ software engineer with 15+ years across embedded Linux, HPC, and Qt application development, and a strong mathematics background. I deliver and optimise production code at every level of the stack — from kernel drivers and Yocto-based BSPs, through Qt 5/6 desktop and instrument-control applications, to OpenMP/MPI/CUDA simulation codes running on clusters. Track record of measurable performance gains (2x–5500x on real workloads), diagnosing complex low-level issues, and of delivering in Agile/SAFe environments alongside multinational teams. Trilingual: Greek (native), English (fluent), French (DELF B2).

WORK EXPERIENCE

Software Engineer (CDI)

Jun 2025 – Present

Analog Way (*audiovisual hardware manufacturer*)

C++, Qt and embedded Linux development on professional video equipment.

- Develop backend services and Linux-side glue connecting web-based control UIs to microcontroller subsystems.
- Containerised existing applications with Docker and modernised CI/CD pipelines to run on container-based runners.
- Contribute fixes and new features across the product line.
- Embedded Linux work on Yocto / BSP-based systems.

C++17, Qt, Yocto, Linux kernel, device tree, Docker, CI/CD, GitLab

Software Engineer (CDI)

Mar 2021 – Jun 2025

Thermo Fisher Scientific (*scientific instrumentation*)

Contributed to Velox, a C++/Qt application controlling electron microscopes, as part of a distributed team spanning several countries.

- Worked across both UI (Qt views, controls, widgets) and backend (instrument communication, data flow) of a complex scientific-instrument control application.
- Worked within a heavily templated, modern C++17 codebase using established GUI architecture patterns (MVP/MVC/MVVM family).
- Delivered features and fixes within an Agile/SAFe release-train environment alongside distributed teams.

C++17, Qt, GUI architecture patterns, Agile, SAFe

C++ Developer (CDI)

Nov 2020 – Mar 2021

FORTHNET (*Greek telecoms / IPTV / energy*)

Maintained and extended the in-house C++ billing system for a major Greek telecom operator.

- Bug fixes and small features on a multi-threaded C++ / Oracle SQL billing system.

C/C++, Linux, Oracle SQL, multithreading

EDUCATION

MSc, High Performance Computing

University of Edinburgh / EPCC, 2008

Thesis: "Derived Metrics with Paraver using hardware counters on Power5 chips".

BSc, Applied Mathematics

University of Crete, 2007

Numerical analysis, linear algebra, solid and fluid mechanics, computer science.

MSc, Electronic & Computer Engineering (*incomplete*)

Technical University of Crete (TUC), 2008

Completed coursework toward an MSc at TUC (3D computer graphics, GPGPU/CUDA, machine learning) but did not submit the dissertation; degree not held. Course projects included a CUDA-accelerated GIS algorithm in MATLAB and an oil-reservoir simulation ported from MATLAB to multi-threaded C (~8x on dual quad-core) and then to CUDA (~5.5x further on a GTX 280).

Software Engineer (CDD)

INRIA (*applied mathematics research laboratory*)

Contributed to ParMMG, the MPI-parallel version of the open-source MMG mesh-adaptation library (mmgtools.org), at INRIA's applied-mathematics laboratory.

- C/C++ development on the parallel re-meshing algorithms and surrounding infrastructure.
- Bug fixes, automated unit-test development, and maintenance of the project's Jenkins/Apache CloudStack CI system.

C, C++, MPI, OpenMP, CMake, Git, Jenkins, Apache CloudStack

Software Engineer (CDD)

INRIA (*applied mathematics research laboratory*)

Performance and reliability work on multiple HPC simulation codes (C, C++/CUDA, Fortran) at INRIA's applied-mathematics laboratory, in collaboration with researchers and PhD students.

- Algorithmic redesign delivered up to **5500x speedup** on individual compute kernels, with **2x-4x** end-to-end gains across whole simulations.
- **Reduced** simulation **memory** footprint by **20-50%** via **data-structure optimisation**.
- Bug fixes, automated unit-test development, and maintenance of the lab's CI infrastructure (Jenkins / Apache CloudStack).
- Operated jobs on Slurm-managed Linux clusters; orchestration and post-processing in bash and Python.

C, C++, CUDA, Fortran, MPI, OpenMP, Jenkins, Apache CloudStack, Slurm, HPC

Software Engineer (CDD)

CNRS (*astrophysics research laboratory*)

Performance and parallelisation work on a production C++/Fortran atmospheric simulation code (MPI + OpenMP) at a CNRS astrophysics laboratory.

- Profiled and optimised hot paths in a long-running simulation codebase, improving its scalability for new simulation campaigns.
- Ran and managed large-scale jobs on Slurm-managed Linux clusters; automated pre/post-processing pipelines in bash and Python.

C++11, Fortran, OpenMP, MPI, Slurm, Bash, Python, HPC

Software Engineer (CDI)

AKKA Technologies (*consulting*)

Profiled and optimised an Unreal Engine 4 automotive HMI/cockpit visualisation scene to lift its frame rate.

- GPU-side performance work across the rendering pipeline — draw-call profiling, render-pass analysis, shader-cost reduction — driven by bottleneck analysis at both engine and project-blueprint level.
- Nearly **2x frame rate**, reaching the limit set by the project's blueprints.

Unreal Engine 4, C++, GPU profiling, real-time rendering

Mar 2017 – Sep 2018

Mar 2015 – Mar 2017

Nov 2019 – Nov 2020

Oct 2018 – Dec 2018

System Administrator & Operations Engineer (CDD)

Nov 2013 – Mar 2015

HCMR (*biology research institute*)

Sole administrator of HCMR's HPC Linux cluster supporting biology and genetics research; also operations support for the institute's web developers.

- Owned the cluster end-to-end: designed and led its upgrade to 600 cores, InfiniBand interconnect, and a ~200 TB Lustre parallel file system.
- Built an automated multi-server backup system based on BACULA.
- Provided dev/ops support to web developers across a stack including Drupal, Tomcat, PostgreSQL, R, and QEMU/Virtual VMs — including for the X-ray 3D image-reconstruction component of lifewatchgreece.eu.
- Implemented an internal C++/CUDA/Qt utility (using GMP for arbitrary-precision arithmetic) computing biological indices from CSV input.

Linux, HPC, Lustre, InfiniBand, BACULA, QEMU, VirtualBox, Drupal, Tomcat, PostgreSQL, R, C++, CUDA, Qt, GMP

Software Engineer (CDI)

Feb 2012 – Jul 2013

Imagination Technologies (*semiconductor design*)

Kernel-mode driver and firmware development on the PowerVR 3D graphics IP, working within Imagination's GPU driver team.

- Bug fixes and feature work across kernel-mode driver (KMD) code and on-GPU firmware.
- Software-side bring-up and validation against simulators and dev boards.
- Multi-platform targets: Android, Linux, and Windows Mobile.

C, Linux kernel, KMD, GPU firmware, Android, Windows Mobile, Perforce

Software Engineer (freelance)

Feb 2009 – Nov 2009

HCMR (*biology research institute*)

Parallelised an existing Fortran 77 ocean-wave simulation code at HCMR using OpenMP, after a previous OpenMP attempt by the client (a domain scientist) had failed to yield any speed-up.

- Worked closely with the client to diagnose why the prior parallelisation had not paid off, then profiled the serial baseline and added effective OpenMP parallelisation to the compute-heavy loops.
- ~**2x speed-up** on SGI Itanium 2; ~**3x–7x** on dual quad-core Intel workstations.

Fortran 77, OpenMP, performance profiling

Undergraduate Research Intern (stage)

2005 – 2007

IACM / FORTH (*applied mathematics research institute*)

3D mesh generation and CFD/biomechanics simulations during BSc studies, on geometries ranging from helicopter CAD to patient arteries (CT-scan).

ANSA, Gambit, Fluent, Abaqus, CFD

KEYWORDS

C, C++, CMake, MPI, OpenMP, CUDA, SyCL, Linux, kernel, embedded, Yocto, BSP, cluster, Slurm, gdb, valgrind, gprof, perf, oprofile, clang-analyzer, Git, GitLab, Subversion, Perforce, Jenkins, Apache CloudStack, QEMU, VirtualBox, BLAS, LAPACK, Eigen, Intel MKL, HDF5, MMG / ParMMG, 3D, CAD, CAE, Qt, MATLAB, Python, Fortran, Go, C#, Agile, SCRUM, SAFe.